

Assignment-4

(20251401022)

NVP

DVP

(Q.1) Find the Fourier Transform of the function

$$f(x) = \begin{cases} 1 & , |x| \leq 1 \\ 0 & , |x| > 1 \end{cases}$$

Also graph $f(x)$.

$$\text{Given, } f(x) = \begin{cases} 1 & , |x| \leq 1 \\ 0 & , |x| > 1 \end{cases} \quad \text{--- (1)}$$

$$\text{Here } f(-x) = f(x)$$

$\therefore f(x)$ is even function

The formula for Cosine Transform is,

$$F_c(\lambda) = \int_0^{\infty} f(u) \cos \lambda u du$$

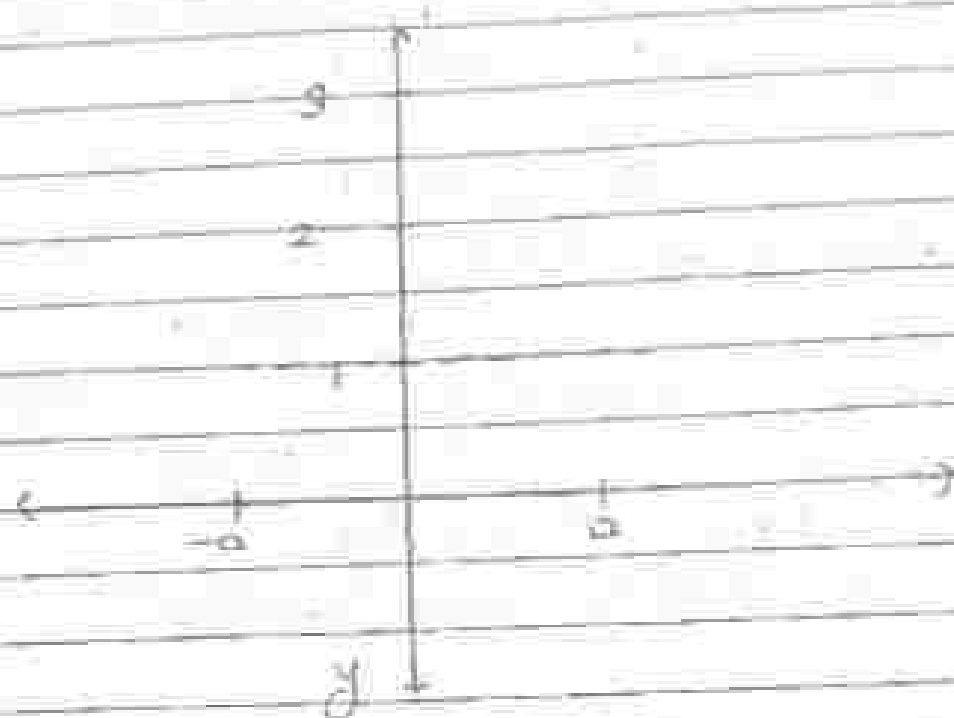
$$= \int_0^a f(u) \cos \lambda u du + \int_a^{\infty} f(u) \cos \lambda u du$$

$$= \int_0^a \cos \lambda u du + \int_a^{\infty} (0) \cos \lambda u du$$

$$= \left[\frac{\sin \lambda u}{\lambda} \right]_0^a$$

$$= \left[\frac{\sin \lambda a}{\lambda} - \frac{\sin \lambda(0)}{\lambda} \right]$$

$$\therefore F_c(\lambda) = \frac{\sin \lambda a}{\lambda}$$



(Q.2) Find the Fourier Transform of the function

$$f(x) = \begin{cases} 1-x^2 & , |x| \leq 1 \\ 0 & , |x| > 1 \end{cases}$$

Given,

$$f(x) = \begin{cases} 1-x^2 & , |x| \leq 1 \\ 0 & , |x| > 1 \end{cases} \quad \text{--- ①}$$

Here, $f(x) = f(-x)$

$\therefore f(x)$ is even function

The formula for Cosine Transform is,

$$\begin{aligned} F_c(\lambda) &= \int_0^{\infty} f(u) \cos \lambda u \, du \\ &= \int_0^1 f(u) \cos \lambda u \, du + \int_1^{\infty} f(u) \cos \lambda u \, du \\ &= \int_0^1 (1-u^2) \cos \lambda u \, du + \int_1^{\infty} (0) \cos \lambda u \, du \end{aligned}$$

$$= \int_0^1 (1-u^2) \cos \lambda u \, du$$

$$= \left\{ (1-u^2) \left(\frac{\sin \lambda u}{\lambda} \right) - (-2u) \left(\frac{-\cos \lambda u}{\lambda^2} \right) + (-2) \left(\frac{-\sin \lambda u}{\lambda^3} \right) \right\}_0^1$$

$$= \left\{ (1-u^2) \frac{\sin \lambda u}{\lambda} - \frac{2u \cos \lambda u}{\lambda^2} + \frac{2 \sin \lambda u}{\lambda^3} \right\}_0^1$$

$$= \left\{ \left[(1-1) \frac{\sin \lambda}{\lambda} - \frac{2(1) \cos \lambda}{\lambda^2} + \frac{2 \sin \lambda}{\lambda^3} \right] - \left[(1-0) \frac{\sin(0)}{\lambda} - 0 + 0 \right] \right\}$$

$$= \left\{ \left[0 - \frac{2 \cos \lambda}{\lambda^2} + \frac{2 \sin \lambda}{\lambda^3} \right] - [0 - 0 + 0] \right\}$$

$$\therefore F_c(\lambda) = \frac{2 \sin \lambda}{\lambda^3} - \frac{2 \cos \lambda}{\lambda^2}$$

(Q3) Find the Fourier Transform of the function $f(x) = e^{-|x|}$

$$f(x) = e^{-|x|} \quad \text{--- (1)}$$

$$|x| \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases} \quad \text{--- (2)}$$

$$\text{Here, } f(-x) = f(x)$$

$\therefore f(x)$ is even function

The Fourier T. for Cosine Transform is,

$$F_c(\lambda) = \int_0^{\infty} f(u) \cos \lambda u \, du$$

$$= \int_0^{\infty} e^{-u} \cos \lambda u du$$

$$= \left\{ \frac{e^{-u}}{(1+\lambda^2)\lambda^2} [-\cos \lambda u - \lambda \sin \lambda u] \right\}_0^{\infty}$$

$$= \left\{ \left[\frac{e^{-\infty}}{1+\lambda^2} (-\cos(\infty) - \lambda \sin(\infty)) \right] - \left[\frac{e^0}{1+\lambda^2} (-\cos(0) - \lambda \sin(0)) \right] \right\}$$

$$= \left\{ [0] - \left[\frac{1}{1+\lambda^2} (-1-0) \right] \right\}$$

$$F_c(\lambda) = \frac{1}{1+\lambda^2}$$

Sine

(Q.6) Find the Fourier Transform of the function $f(x) = \frac{1}{x}$

Given, $f(x) = \frac{1}{x}$ — (1)

The formula for Fourier Sine Transform is,

$$F_s(\lambda) = \int_0^{\infty} f(u) \sin \lambda u du$$

$$= \int_0^{\infty} \frac{1}{u} \sin \lambda u du$$

Put $\lambda u = t$

$$u = \frac{t}{\lambda}$$

Lower limit $u=0$; $t=0$ $\left[\because u = \frac{t}{\lambda} \right]$

Upper limit $u=\infty$; $t=\infty$

$$= \int_0^{\infty} \lambda \sin t \frac{dt}{\lambda}$$

$$= \int_0^{\infty} \sin t \, dt$$

$$\left[\frac{\sin x}{x} \, dx = \begin{bmatrix} \pi/2 & 000 \\ -\pi/2 & 000 \end{bmatrix} \right]$$

$$\therefore F_3(\lambda) = \frac{\pi}{2}$$

(Q5) Solve the following integral equation.

$$\int_0^{\infty} f(x) \sin \lambda x \, dx = \begin{cases} 1-\lambda & 0 \leq \lambda < 1 \\ 0 & \lambda \geq 1 \end{cases}$$

Here,

$$f(\lambda) = \begin{cases} 1-\lambda & 0 \leq \lambda < 1 \\ 0 & \lambda \geq 1 \end{cases} \quad \text{--- (1)}$$

The formula for Inverse Sine Transform is,

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F(s) \lambda \sin \lambda x \, d\lambda$$

$$= \frac{2}{\pi} \left\{ \left[\int_0^1 F(s) \lambda \sin \lambda x \, d\lambda + \int_1^{\infty} F(s) \lambda \sin \lambda x \, d\lambda \right] \right\}$$

$$= \frac{2}{\pi} \left\{ \int_0^1 (1-\lambda) \sin \lambda x \, d\lambda + \int_1^{\infty} (0) \sin \lambda x \, d\lambda \right\}$$

$$= \frac{2}{\pi} \left\{ (1-\lambda) \left(-\frac{\cos \lambda x}{\lambda} \right) - (-1) \left(-\frac{\sin(\lambda x)}{\lambda^2} \right) \right\}_0^1$$

$$= \frac{2}{\pi} \left\{ -\frac{(1-\lambda) \cos \lambda x}{\lambda} - \frac{\sin(\lambda x)}{\lambda^2} \right\}_0^1$$

$$= \frac{2}{\pi} \left\{ \left[-\frac{(1-1) \cos x}{1} - \frac{\sin(x)}{1^2} \right] - \left[-\frac{(1-0) \cos(0)}{1} - \frac{\sin(0)}{1^2} \right] \right\}$$

$$= \frac{2}{\pi} \left\{ \left[0 - \frac{\sin x}{1^2} \right] - \left[-1 - 0 \right] \right\}$$

$$\therefore f(x) = \frac{2}{\pi} \left\{ \frac{-\sin x}{1^2} + \frac{1}{1} \right\}$$

(Q.6) Solve the following integral equation.

$$\int_0^{\infty} f(x) \cos \lambda x dx = e^{-\lambda} \quad ; \quad \lambda > 0$$

$$\text{Here } f_c(\lambda) = e^{-\lambda} \quad ; \quad \lambda > 0 \quad \text{--- (1)}$$

The formula for Inverse Cosine Transform is,

$$f(x) = \frac{2}{\pi} \int_0^{\infty} f_c(\lambda) \cos \lambda x d\lambda$$

$$f(x) = \frac{2}{\pi} \int_0^{\infty} e^{-\lambda} \cos \lambda x d\lambda \quad \begin{matrix} a = (-1) \\ b = (x) \end{matrix}$$

$$= \frac{2}{\pi} \left\{ \frac{e^{-\lambda}}{(-1)^2 + x^2} (-\cos \lambda x + x \sin \lambda x) \right\}_0^{\infty}$$

$$= \frac{0}{\pi} \left\{ \left[\frac{e^{-m}}{1+m^2} (-\cos(m)) + \pi \sin(m) \right] - \left[\frac{e^0}{1+0^2} (-\cos(0) + \pi \sin(0)) \right] \right\}$$

$$= \frac{0}{\pi} \left\{ [0] - \left[\frac{1}{1+0^2} (-1) \right] \right\}$$

$$\therefore \boxed{f(m) = \frac{2}{\pi} \left(\frac{1}{1+m^2} \right)}$$

(Q7) Solve the following integral eqⁿ.

$$\int_0^{\infty} f(\lambda) \sin \lambda x d\lambda = \begin{cases} 1 & 0 \leq \lambda < 1 \\ 2 & 1 \leq \lambda < 2 \\ 0 & \lambda \geq 2 \end{cases}$$

Given, $f_s(\lambda) = \begin{cases} 1 & 0 \leq \lambda < 1 \\ 2 & 1 \leq \lambda < 2 \\ 0 & \lambda \geq 2 \end{cases}$

The formula for Inverse Sine Transform is,

$$f(x) = \frac{2}{\pi} \int_0^{\infty} f_s(\lambda) \sin \lambda x d\lambda$$

$$= \frac{2}{\pi} \left\{ \int_0^1 f(\lambda) \sin \lambda x d\lambda + \int_1^2 f(\lambda) \sin \lambda x d\lambda + \int_2^{\infty} f(\lambda) \sin \lambda x d\lambda \right\}$$

$$= \frac{2}{\pi} \left\{ \int_0^1 \sin \lambda x d\lambda + \int_1^2 2 \sin \lambda x d\lambda + \int_2^{\infty} (0) \sin \lambda x d\lambda \right\}$$

$$= \frac{2}{\pi} \left\{ \left[\frac{-\cos \lambda x}{x} \right]_0^1 + \left[\frac{-2 \cos \lambda x}{x} + \left(\frac{0 (\sin(0))}{x} \right) \right]_2^{\infty} \right\}$$

$$= \frac{2}{\pi} \left\{ \left[\frac{-\cos x}{x} + \frac{\cos(0)}{x} \right] + \left[\frac{-2 \cos(2x)}{x} + \frac{2 \cos x}{x} \right] \right\}$$

$$= \frac{2}{\pi} \left\{ \left(\frac{1 - \cos x}{x} \right) - 2 \left(\frac{\cos 2x - \cos x}{x} \right) \right\}$$

$$= \frac{2}{\pi x} \left\{ 1 - \cos x - 2 \cos 2x + 2 \cos x \right\}$$

$$= \frac{2}{\pi x} \left\{ \cos x - 2 \cos 2x + 1 \right\}$$

$$\therefore f(x) = \frac{2}{\pi x} (1 + \cos x - 2 \cos 2x)$$

(Q8) Find the Fourier Integral representation for the function,

$$f(x) = \begin{cases} x^2, & 0 \leq x < a \\ 0, & x \geq a \end{cases}$$

Given, $f(x) = \begin{cases} x^2, & 0 \leq x < a \\ 0, & x \geq a \end{cases} \quad (\because x^2 \rightarrow \text{even function})$

The formula for Fourier Cosine Transform is,

$$F_c(\lambda) = \int_0^{\infty} f(u) \cos \lambda u \, du$$

$$= \int_0^a f(u) \cos \lambda u \, du + \int_a^{\infty} (0) \cos \lambda u \, du$$

$$= \int_0^a u^2 \cos \lambda u \, du$$

$$= \left\{ u^2 \left(\frac{\sin \lambda u}{\lambda} \right) - 2u \left(-\frac{\cos \lambda u}{\lambda^2} \right) + 2 \left(-\frac{\sin \lambda u}{\lambda^3} \right) \right\}_0^a$$

$$= \left\{ \frac{u^2 \sin \lambda u}{\lambda} + \frac{2u \cos \lambda u}{\lambda^2} - \frac{2 \sin \lambda u}{\lambda^3} \right\}_0^a$$

$$= \left\{ \left[\frac{a^2 \sin \lambda a}{\lambda} + \frac{2a \cos \lambda a}{\lambda^2} - \frac{2 \sin \lambda a}{\lambda^3} \right] - \left[0 + 0 - \frac{2 \sin(0)}{\lambda^3} \right] \right\}$$

$$F_c(\lambda) = \frac{a^2 \sin \lambda a}{\lambda} + \frac{2a \cos \lambda a}{\lambda^2} - \frac{2 \sin \lambda a}{\lambda^3}$$

$\therefore$ The formula for Inverse Fourier Cosine Transform is $\rightarrow$

$$f(x) = \frac{2}{\pi} \int_0^{\infty} f_c(\lambda) \cos \lambda x d\lambda$$

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \left[\frac{a^2 \sin \lambda a}{\lambda} + \frac{2a \cos \lambda a}{\lambda^2} - \frac{2 \sin \lambda a}{\lambda^3} \right] \cos \lambda x d\lambda$$

(Q.9) Using Fourier Sine Integral representation, Show that

$$\int_0^{\infty} \frac{\lambda \sin \lambda x}{\lambda^2 + m^2} d\lambda = \frac{\pi}{2} e^{-mx}, \quad m > 0, x > 0$$

Here, $f(x) = \frac{\pi}{2} e^{-mx}, \quad m > 0, x > 0$ — (1)

The formula for Fourier Sine Transform is

$$f_s(\lambda) = \int_0^{\infty} f(u) \sin(\lambda u) du$$

$$= \int_0^{\infty} \frac{\pi}{2} e^{-mu} \sin(\lambda u) du$$

$$= \frac{\pi}{2} \int_0^{\infty} e^{-mu} \sin(\lambda u) du \quad \begin{matrix} a = (-m) \\ b = (\lambda) \end{matrix}$$

$$= \frac{\pi}{2} \left[\frac{e^{-mu}}{(-m^2) + (\lambda^2)} (-m \sin(\lambda u) - \lambda \cos(\lambda u)) \right]_0^{\infty}$$

$$= \frac{\pi}{2} \left\{ \left[\frac{e^{-m\omega}}{m^2 + \lambda^2} (-m \sin(\omega) - \lambda \cos(\omega)) \right] - \left[\frac{e^0}{m^2 + \lambda^2} (-m \sin(0) - \lambda \cos(0)) \right] \right\}$$

$$= \frac{\pi}{2} \left\{ \left[0 - \left[\frac{1}{m^2 + \lambda^2} \right] \lambda \right] \right\}$$

$$= \frac{F(s)\lambda}{2(m^2 + \lambda^2)} = \frac{\pi\lambda}{2(m^2 + \lambda^2)} \quad \text{--- (2)}$$

The formula for Fourier Inverse Sine Transform is

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F(s) \lambda \sin \lambda x \, d\lambda$$

$$\therefore f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\pi}{2} \left(\frac{\lambda}{m^2 + \lambda^2} \right) \sin \lambda x \, d\lambda$$

$$\therefore \frac{\pi}{2} e^{-mx} = \int_0^{\infty} \frac{\lambda \sin \lambda x}{m^2 + \lambda^2} \, d\lambda$$

(Q.10) Using Fourier Sine Integral representation, Show that

$$\int_0^{\infty} \frac{\cos \lambda x}{\lambda^2 + m^2} \, d\lambda = \frac{\pi}{2m} e^{-mx} \quad ; \quad m > 0, \quad x > 0$$

Here,

$$f(x) = \frac{\pi}{2m} e^{-mx} \quad ; \quad m > 0 \neq x > 0$$

The formula

$$f_c(\lambda) = \int_0^{\infty} f(u) \cos(\lambda u) \, du$$

$$= \frac{\pi}{2m} \int_0^{\infty} e^{-mu} \cos(\lambda u) du$$

$$= \frac{\pi}{2m} \left\{ \left[\frac{e^{-mu}}{(-m)^2 + \lambda^2} (-m \cos(\lambda u) + \lambda \sin(\lambda u)) \right] \right\}_0^{\infty}$$

$$= \frac{\pi}{2m} \left\{ \left[\frac{e^{-\infty}}{m^2 + \lambda^2} (-m \cos(\infty) + \lambda \sin(\infty)) \right] - \left[\frac{e^0}{m^2 + \lambda^2} (-m \cos(0) + 0) \right] \right\}$$

$$= \frac{\pi}{2m} \left\{ [0] - \left[\frac{1}{m^2 + \lambda^2} (-m) \right] \right\}$$

$$= \frac{\pi (m)}{2m(m^2 + \lambda^2)}$$

$$F_c(\lambda) = \frac{\pi}{2(m^2 + \lambda^2)} \quad \text{--- (2)}$$

The formula for Inverse Cosine Transform is,

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F_c(\lambda) \cos(\lambda x) d\lambda$$

$$\therefore f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\pi}{2} \left(\frac{1}{m^2 + \lambda^2} \right) \cos(\lambda x) d\lambda$$

$$\therefore f(x) = \int_0^{\infty} \frac{1}{m^2 + \lambda^2} \cos \lambda x d\lambda$$

$$f(x) = \frac{\pi}{2m} e^{-mx} = \int_0^{\infty} \frac{\cos \lambda x}{m^2 + \lambda^2} d\lambda$$

(Q.11) Given.

$$f(x) = \begin{cases} \pi/2 \sin x, & |x| \leq \pi \\ 0, & |x| > \pi \end{cases}$$

Put $x = -x$

$$f(-x) = \begin{cases} \pi/2 \sin(-x), & |-x| \leq \pi \\ 0, & |-x| > \pi \end{cases}$$

$$\therefore f(-x) = -f(x)$$

 $f(x)$ is Odd function

The formula for Fourier Sine Transform is,

$$f_s(\lambda) = \int_0^{\infty} f(u) \sin(\lambda u) du$$

$$= \int_0^{\pi} f(u) \sin(\lambda u) du + \int_{\pi}^{\infty} f(u) \sin(\lambda u) du$$

$$= \int_0^{\pi} \pi/2 \sin(u) \sin(\lambda u) du + \int_{\pi}^{\infty} (0) \sin(\lambda u) du$$

$$= \frac{\pi}{2} \int_0^{\pi} \sin(u) \sin(\lambda u) du$$

As we know, $\left(\sin(A) \sin(B) = \frac{1}{2} (\cos(A-B) - \cos(A+B)) \right)$

$$\therefore \frac{\pi}{2} \int_0^{\pi} \frac{1}{2} (\cos(u-\lambda u) - \cos(u+\lambda u)) du$$

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$$= \frac{\pi}{4} \int_0^{\pi} \cos(u-\lambda u) - \cos(u+\lambda u) du$$

$$= \frac{\pi}{4} \left\{ \frac{\sin(1-\lambda)u}{(1-\lambda)} - \frac{\sin(1+\lambda)u}{(1+\lambda)} \right\}_0^{\pi}$$

$$= \frac{\pi}{4} \left\{ \left[\frac{\sin(1-\lambda)\pi}{(1-\lambda)} - \frac{\sin(1+\lambda)\pi}{(1+\lambda)} \right] - \left[\frac{\sin(1-\lambda)0}{1-\lambda} - \frac{\sin(1+\lambda)0}{(1+\lambda)} \right] \right\}$$

$$= \frac{\pi}{4} \left\{ \frac{\sin(1-\lambda)\pi}{(1-\lambda)} - \frac{\sin(1+\lambda)\pi}{(1+\lambda)} \right\}$$

sin(b

$$F_S(\lambda) = \frac{\pi}{4} \left\{ \frac{\sin(1-\lambda)\pi}{(1-\lambda)} - \frac{\sin(1+\lambda)\pi}{(1+\lambda)} \right\}$$

The formula for inverse Sine Transform is,

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F_S(\lambda) \sin \lambda x d\lambda$$

$$f(x) = \frac{2}{\pi} \times \frac{\pi}{4} \int_0^{\infty} \left\{ \frac{\sin(1-\lambda)\pi}{(1-\lambda)} - \frac{\sin(1+\lambda)\pi}{(1+\lambda)} \right\} \sin \lambda x d\lambda$$

$$f(x) = \frac{1}{2} \int_0^{\infty} \left\{ \frac{\sin(1-\lambda)\pi}{(1-\lambda)} - \frac{\sin(1+\lambda)\pi}{(1+\lambda)} \right\} \sin \lambda x d\lambda$$

(Q.12) Given, $f(x) = e^{-|x|}$ $-\infty < x < \infty$ — (1)

$$f(x) = f(-x)$$

$\therefore f(x)$ is even function.

Formula for Fourier Cosine Transform is,

$$F_c(\lambda) = \int_0^{\infty} f(u) \cos(\lambda u) du$$

$$F_c(\lambda) = \int_0^{\infty} e^{-u} \cos \lambda u du$$

By using formula, $\int e^{at} \cos(bt) dt = \frac{e^{at}}{a^2 + b^2} (a \cos(bt) + b \sin(bt))$

$$\therefore F_c(\lambda) = \left[\frac{e^{-u}}{1 + \lambda^2} (-\cos(\lambda u) + \lambda \sin(\lambda u)) \right]_0^{\infty}$$

$$= \left\{ \left[\frac{e^{\infty}}{1 + \lambda^2} (-\cos(\infty) + \lambda \sin(\infty)) \right] - \left[\frac{e^0}{1 + \lambda^2} (-\cos(0) + \lambda \sin(0)) \right] \right\}$$

$$= \left\{ [0] - \left[\frac{1}{1 + \lambda^2} (-1) \right] \right\}$$

$$F_c(\lambda) = \frac{1}{(1 + \lambda^2)} \quad \text{--- (2)}$$

Formula for Inverse Cosine Transform is,

$$f(x) = \frac{2}{\pi} \int_0^{\infty} F_c(\lambda) \cos \lambda x d\lambda$$

$$\therefore f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\cos(\lambda x)}{1+\lambda^2} d\lambda$$